

0 5

Scientists investigated the role of a protein called CENP-W in mitosis. Their method involved cell fractionation and ultracentrifugation.

0 5 . 1

The scientists began by lysing (breaking open) cells and organelles using a detergent that dissolves lipids in water.

Suggest how the detergent releases CENP-W from cells.

[2 marks]

0 5 . 2

Explain how ultracentrifugation separates CENP-W from other molecules.

[2 marks]

The scientists treated cells in a culture with small interfering RNA (siRNA). This siRNA causes RNA interference of expression of the *CENP-W* gene. The scientists took samples of cells at 0, 48 and 72 hours after adding the siRNA. They then used gel electrophoresis to separate CENP-W and tubulin from these samples.

Figure 5



[3 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

